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the integrated Man of tomorrow

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*** START OF THE PROJECT GUTENBERG EBOOK CORBOW'S
THEORY ***

Corbow's Theory

BY LEE WALLOT

*It was a terrific theory and it would send
Man to the stars. But the two men involved had
to buck more than physical laws; and so the
project was finished, over, done with. Unless....*

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"All right! So we've got it. The same problem rocket designers have been struggling with for five years. Nobody's found the answer—and they never will!"

Bronsen Corbow glared at the older man, his lips pressed tightly together to keep from giving voice to the anger mounting inside of him. Mars Kenton was an argumentative old fool, but the company had made him his assistant and nothing could be done about it.

"They've known ever since they discovered that interstellar drive," Mars continued, "that they can only make enough Carbolium to send four ships a year to the end of our galaxy and back again. Is it our fault they have to make the blasted stuff instead of mining it out of the ground?"

The words ringing in the quiet of the laboratory seemed to pound in Bronsen's ears and he found he could hold his tongue no longer. He leaned toward the older physicist and slammed his hand down on the table.

"That's enough, Mars. I happen to be the one in charge here, not you." His quiet voice made clear the anger he felt. "Reed turned the problem over to us. I say we can lick it. Just because my chief assistant is still thinking in terms of ancient history, it's no reason to send back a report from this laboratory saying we can't handle the problem." He ran a trembling hand through his close-cropped hair and swore at himself when he saw Mars noticed the trembling. Why did he have to start shaking every time he got mad? The person he was mad at invariably took the shaking to be fear, and he would always be forced to drive his point home all the harder in order to get the respect he demanded.

Mars Kenton sneered. "Mind telling me just how you are going to eliminate interstellar drive from our rocket ships? Or have you cooked up another of your bright ideas to try out at the company's expense?"

"I'm fed up with you, Mars!" All control over his temper was gone now and the younger man gave full vent to his anger. His powerful body fairly bristled in his rage and in spite of himself Mars was forced to cringe beneath the assailing roars that followed. "You may be twenty years older than I am; you may have been one of the pioneers in space travel; you may

still be a good man if you could forget that the whole world didn't plot that accident that left you with a bad leg—but you're still taking orders from me. We have some good men in this department, and you can either keep your mouth shut and work with us or you can get out. Interstellar drive isn't the only solution to space travel and the answer to the problem is going to come from this laboratory. Now take your choice!"

Mars glared at Bronsen and seethed inwardly but swung back to his work table. His right leg twitched convulsively, forcing him into a stumbling limp and he silently cursed the fate that had brought him to such a lowly existence. Him! Joc Kenton! Member of the first expedition to land on Mars and successfully return to Earth. And what was he now? Just a second rate design consultant working in a laboratory on the moon. His water blue eyes clouded in his flood of self pity. How beautiful it had been out there ... all blackness, all majesty, the throbbing power of the rockets, the thrill of unknown adventures in the void. His rickety old heart beat faster with remembering. The scorching desolateness of Mars was something he would never forget. Even now he could see the miles of heat-drenched land, the thick red powder that covered the planet's crust, the stretching reaches of nothing but a barren, dead world. And then—the accident. Sure, it was just an accident. How could he know that the port lid was going to break its magnetic field and slam down upon him? It had though, and he had returned an honored man, praised for his self-sacrificing adventure, then pitied because he would spend the rest of his life a crippled man. He twisted his thin, blue-veined hands together, those hands that had piloted a glittering rocket through space, those hands that had sifted through the sands of an alien world, those hands that now were white and fragile, working over drawings and plans for other ships. Gone were the dreams, and with their going came the bitterness.

He felt his anger melting in his own self pity, decided not to brush away the tears that gathered in his eyes and turned to his board, staring at it through blurred vision.

"Bong! End of round five. Just wait around a minute folks, next round coming up." Vern Webber peered cautiously around the door as if expecting something to fly at him, then jumped into the room. His youthful face broke into a broad grin as he bowed before the chief designer. "Oh great and noble Mr. Bronsen Corbow. Is it safe for your lowly servant to approach these

hallowed halls in answer to your summons? Mine is not to reason why—but I'd still like to leave here with my head on my shoulders."

Bronsen found himself smiling at his young assistant. Vern, although he was twenty-four, had the spirit and air of a teenager and usually succeeded in keeping the lab in a state of high humor. The tenseness of the argument with Mars dispelled itself and Bronsen relaxed.

"Get word to the men that we are having a special meeting this afternoon, in the conference room. We're going to blow the lid right off the present concepts of space travel and really give those people out there something that will make their eyes bulge. I'll tell you more about it this afternoon."

"Aye, aye, sir!" Vern clicked his heels, gave an exaggerated salute and was gone.

Bronsen glanced in returning annoyance at the snort of disgust that issued from Mars' corner. That old fool and his rockets, he thought were things of the past. There was only the future now. New ideas, new methods, new successes. Why couldn't Mars see that? And yet, Bronsen himself felt a tiny pulsing of doubt. He cursed himself for that tinge of self-distrust, but could do nothing about it. He was brilliant, he was a master of design and he knew space flight as well as he knew the shape, workings and complexities of the pencil he twirled in his hands. But what if he wasn't right? What if his new theory was a flop—and with it a waste of money, time and human lives?

He tore himself from his dismal thoughts with a savage determination and strode into his office. That damned Mars was just getting under his skin, that's all. Listen to him long enough and he'll have you thinking we should all have stayed back in the horse and buggy days. No problems of space flight then, no old beliefs, ancient ideas, stagnant prejudices to worry about then. Not as far as traveling to the stars was concerned, anyway.

"First, let's review a few points I know you are all familiar with, but that you should keep in mind, starting with the beginning of space travel."

Bronsen ran his hand through his hair and looked over the eagerly expectant faces of his staff, considering carefully the points he had to make.

"First, there were the Detorium-driven rockets. Fine ships that opened up the realm of travel in outer space." His voice was firm, stringent with his inner excitement, his faith in his idea. "In fact, it was just such a ship as this that Joc Kenton, Mars to you folks, was on when they made their first landing on Mars. These ships had their good points and still are excellent over short distances but certainly are of no use for intergalactic flight. They are much too slow, requiring more than a lifetime to make the trip there and back.

"Then came the interstellar drive, the method we are now using. This drive, utilizing Carbolium which we have to manufacture, makes full use of space as a medium of travel. There's only one catch. We can make only a certain amount of Carbolium, and the costs of making it are astronomical. However, no matter what the expense, the supply is limited. Still, everything's fine. We use the old rockets for short distances and the interstellar drive for intergalactic trips."

He paused a minute to let it sink in. "That is everything *was* fine. As you know, we have found another planet, remarkably similar to Earth in the Eastrex Galaxy, and have succeeded in setting up a colony there. This presents problems. For one, since we can send only four ships a year to Naver, this new planet, our colonizing is going to be slowed down to a crawl. Also, we will have a mighty hard time trying to get supplies and everything else that a new colony needs to Naver within a reasonable length of time. In other words, what we need is a ship that will be cheap to run, fast enough to get there and get back again and also safe enough to carry passengers and cargo, even in small amounts. You've heard of this problem before; I know you've tried independently to work it out and have not had much luck. I think we might have an answer now.

"How many of you have shot a rifle or are familiar with a gun?"

The expectant faces had gone blank. What was he leading up to anyhow? Still, a large number of hands cautiously worked their way into the air.

"All right. All you have to do right now is forget about space ships and concentrate on a rifle and a bullet." There was beginning to be the muttering

and stirring of a confused group of people. Bronsen's muscular hands gripped the table eagerly. They were confused, but they were also interested. "You are familiar with the fields and grooves of a gun barrel," he continued, "and you know they are there to give the bullet a spin when it is fired. Now what happens to a bullet when it is fired from a smooth barrel, with no grooves? It is inaccurate, wobbles, has less power and eventually turns end over end. Now, take the grooved rifle barrel. The bullet is given a spin, it has many times the velocity of the other, it has a straight line accuracy due to the spiral motion—keep these last two points in mind—and providing the rifle is aimed right in the first place, will hit the target."

The group was being split up into two factions, those who leaned forward expectantly once more and those who shook their heads in bafflement.

"Now, let's go back again to the old rockets, and also our present ones. They all use the same principle to get off the ground—blast off with rockets. But let's add a second type of blasting off area, also using rockets, but one that looks like a monstrous rifle barrel, complete to the fields and grooves. We have a launching apparatus that is like a grooved rifle compared to a smooth barreled one. The smooth barrel that we now use, the rockets taking off straight, gives good acceleration but not enough top speed. Our old rockets had that fault and never could get good velocity even when in flight. Our new ones take over with the interstellar drive, but we want to eliminate this last method. The spiral take off though, would give much greater velocity right from the start, would enable the ship to hit outer space with a greater speed than it could attain using the other methods and it would continue on in space at a much faster rate. The lack of friction would keep it from slowing down and if we could hit the speed we want before entering outer space, the ship could go right on at that same rate in space. There wouldn't even be need for rocket power while in actual galactic flight. The initial momentum would carry it through to its destination. The rocket could do all this because this spiral launching would give us more 'muzzle velocity', and in a given time after blast off, the spinning ship would have reached a much greater speed than the regularly fired one. This means operating cost will be confined to blast off, landing and to skirt around any sudden dangers that might arise in space.

"We therefore have a ship that has the velocity of the drive and maybe more, without the cost of drive. It has safety since our old rockets proved to

be remarkably accident-free and this design would actually be working on almost the same principle as the old rockets, except for the blast off. The difference in outer space would be this. The old ships used rockets throughout the trip to increase their velocity. The new ones will be traveling so fast when they enter outer space, they won't have to use any power because they will already have the velocity needed."

He paused to swallow the dryness in his mouth and noticed with pleasure that more than one face was twisted in thought. Good. All they have to do is be given a theory to think about, the time to do that thinking and they'd be on their way.

"That's about it for now. I have some ideas of my own for the design of this ship ... and it's really surprising when you think how simple it is. But I'll save that for some other time. Right now, I'd want all of you to think about it. It gets top priority. Report back to me with your ideas. I think we can lick this problem and get our bread and butter, Inter Galactic Enterprises, right on top of the pile. Good luck."

The meeting broke up amidst frantic discussion, wails of misunderstanding, confusion and quiet self-musings. Bronsen smiled to himself, his face almost boyishly radiant in his pleasure. The seed had been planted. Now all he had to do was give it time to grow and bear fruit.

Vern's cry of amazement rang with reverence as he stood with Bronsen looking over the first test rocket as it was slowly wheeled to the launching area. "Boy-o-boy! Look at that beauty! Did you ever see anything like it?" The silver ship lay on its side as it approached the huge tower, but already Vern could see its glistening majesty soaring through the sky.

"She was a lot of work, Vern. Let's just keep our fingers crossed. By the way, that design of yours on the rotating cylinder inside the rocket, working independently of the rocket's forced spin, was good. Tough thing to lick, but now the pilot can keep a steady 'up' and 'down' no matter how much the outside of the ship is spinning. Good work."

Vern shrugged his shoulders. "Just call me the Einstein of the 23rd century, that's all. We'll all see how well everything works pretty soon now."

Bronsen was finding the tension beginning to build up in him. Just a few more hours and his theory would be lauded with success or shattered into the dust. He peered at the rocket, at the tiny black figures of the men that were dwarfed by its size and at the giant, black tube, towering hundreds of feet high, waiting patiently to receive her first charge. The needle-nosed space craft glistened in the early morning sun, her thin beauty tantalizing to the senses of a spaceman. Her lines swept gracefully back across her smooth expanse until they hit the four fin tips sweeping out from the rotating band of the tail piece, the fin tips that would fit into the slowly spiraling grooves of the launching tower. The field and groove construction first suggested by Bronsen had been replaced by lands and grooves when it was found that the fewer grooves gave greater accuracy and better muzzle velocity when tested on the laboratory models. Thus, there were only four fins instead of the originally planned eight.

The rocket reached the lowering platform and Bronsen watched in nervous anxiety as the ship was lowered into the ground tail first, then slowly began its upward ascent into the belly of the launcher. He thought of a thousand things that could happen right then, found that none of them were going to and returned to his office. The rocket was safely nestled in the launcher's belly, patiently waiting for the human crew to arrive and give it life.

When they gathered about the launching field that afternoon Bronsen found himself sweating in both the heat of the day and the torrid intensity of Mars' insistence that the whole thing was going to be one big flop.

"Just one blast from those rockets and we'll all be blown into the next galaxy, without benefit of a space ship. Trying to shoot a 70,000 pound rocket as if it were a toy gun. You'll learn one of these days, Corbow, that the old way was still the best. It got us to Mars and back and if you'd work on that instead of this, it would be good for intergalactic flight too. But no, you've got to have your name up there in print."

"Oh, shut up your damned mumbling, Mars!"

Bronsen shot the words out savagely. He gave the older man a withering look and turned his attention again to the ship. The men that were to take

her up had disappeared inside the expanse of the launching tower and the other figures darted back and forth, making last minute preparations. The minutes began to tick off. Five minutes until blast off. Four minutes. Three minutes. The field was now completely devoid of human figures. Two minutes. One minute. Ten seconds, nine, eight. The launcher looked lonely and terrifying in its greatness and Bronsen tried to wish the rocket up out of her belly by will power alone. *Four, three, two, one....*

The ground trembled as the ear-shattering roar jumped across the lunar landscape. The sound grew louder, sharper, and Bronsen began to think his head would split with the noise. The rockets pitched higher, their scream pierced the air and then the silver nose of the ship edged above the top of the launcher. It pulled further into view, the shimmering silver glinting in the sun and Bronsen clenched his fists in anxiety. Come on baby; show them what you can do. That's it baby, keep right on coming. Come on girl! The ship rose clear of the launcher, the distance making it look as if it were shot straight out, but Bronsen knew the steadily spinning hull was heading right.

Suddenly he noticed it. Something was wrong! The ship wasn't acting right. What was it? His eyes tried to leap from his head to get closer to the rising needle and then he saw it. It was shaking. The whole ship was trembling as if in human terror. He watched the tremors pass from the nose to the tail, each one more violent than the last, until the whole ship was wracked with a shaking like palsy. Why? It had worked so beautifully with the experimental models. What was causing it? The bow of the ship was now visibly shaking, the tremors becoming more savage and then the nose began to dip. With a final shudder of resignation, the rocket pitched over and began its screaming descent. Bronsen watched the plunging ship, felt his heart grab in pain in his chest and stumbled back from the observation window unable to watch any longer. The burst of a million shells at once slammed into the unyielding lunar plain. In his mind's eye he could see the twisted, exploding mass of metal and the thought sickened him.



The others ran from the room, heading for the wrecked ship. Bronsen watched them with dull eyes and made no attempt to follow. What could they do for the four men that had gone to their deaths in his mad creation? What could they do for the millions of dollars that now lay a twisted heap of rubble? He turned to drag his defeated body back to the lab, to twist and mull in his mind what had happened, and found himself looking into the glaring eyes of Mars.

"I told you, didn't I, Mr. Corbow?"

Bronsen covered his ears so that he wouldn't hear. He screamed, "Shut up! Shut up before I slam you one."

Mars spat in disgust. "Four nice guys in that ship, too. Knew 'em, didn't you?"

Bronsen's hammer-hard fist smashed into Mars' mouth and the old man was slammed against the wall before falling in a crumpled heap on the floor. He sat there, the blood oozing from his mouth as he stared at the retreating back of the man he never thought would have enough nerve to really hit him. Now he was sorry he had said anything and the self pity welled up within him. He really didn't mean half of what he always managed to spit out. What made him do it? He wiped the blood from his mouth and pulled himself to his feet.

Bronsen slumped further down into the soft contours of the chair, eyeing Hanson Reed with a tortured soul. The president of Inter Galactic Enterprises glared at him from the other side of the desk, every inch of his paunchy frame the body of an outraged executive. He chewed violently on the black cigar in his mouth and waited impatiently for Bronsen to explain. Bronsen spread his hands helplessly.

"I don't know Reed. I just don't know." His shoulders heaved in a sigh of dejection. "Every single person in the moon lab has been looking for an answer and we still can't find out why the ship crashed. We've tested the laboratory models over and over again. We've gone over every little detail and have nothing but a blank to show for it."

Reed chewed more savagely at the end of his cold cigar. "We spent two million dollars on research and development and all we have to show for it is a pile of scrap metal and four corpses scattered over the lunar landscape. There's got to be some explanation."

"Just one in a million chances that an accident like this would happen," Bronsen countered desperately. "It's just coincidence that it happened on the first model."

"Coincidence!" Mars' voice was guttural with contempt. "I told you from the start it wasn't practical. I knew..."

"All right, Mars," Reed interrupted. "You were project design engineer, right?" Mars nodded in agreement. "Was there anything wrong with the design of the ship, any reason why it probably wouldn't have worked, from a design stand-point?"

"No," he answered reluctantly. "Not that I could see. I just knew from the start it wasn't going to work. I told Bronsen that, lots of times, but he just isn't the type to take advice."

Bronsen roared and leaped to his feet.

"You old fool," he bellowed. "Technically, theoretically and mechanically, there wasn't one indication that it wasn't going to be a completely successful launching procedure. You know that as well as anyone! Ask the men around you. They handled the final application, the mechanics, the construction, the blast off. Ask any one of them. Every single one of them will tell you the same thing. There was no reason why the ship should have crashed! Every item had been checked, double checked and re-checked again. The instruments indicated everything was functioning perfectly at blast off. If you didn't have such a twisted inverted opinion of everything...."

Mars leaned forward, his body now trembling, "Don't you go calling me names, you swell-headed pup!"

Reed pounded his desk violently.

"Mars! Bronsen!" he shouted impatiently. "This is hardly the time for name-calling and airing personal gripes. We're here to find a good reason for spending more money on this project. We're not children in a schoolyard, arguing over a piece of candy, although that's exactly what it's beginning to sound like. Frankly, I'm of the opinion that with so much internal fighting going on, nothing could possibly come of spending more. It would be a waste of both finances and time."

Bronsen slowly sat down again, his trembling hands clenched into tight fists.

"That's one item you don't have to worry about," he growled. "Kenton is completely finished as far as I'm concerned. He's out. Fired."

Mars' face fell in shocked surprise. Reed tore the cigar from his mouth and glared at Bronsen.

"No one is being fired, Bronsen. You've been a good leader, in my opinion, as well as a friend, but I do the firing around here." Bronsen glowered and reddened under the unexpected rebuttal but said nothing. "You are young yet," Reed continued. "You've got brains, imagination, leadership and ability. Wouldn't be where you are if you didn't. There's just one thing lacking, and Mars is the one that has it. Experience. And with that experience goes well-used caution. You've got the go-ahead, but he has the wisdom. Temperance and drive. That's Mars and you. You've got each other. Why don't you just learn how to work with and use each other?"

Bronsen remained in baleful silence. Mars glared at the younger man and sneered contemptuously.

"That young pup never will know what the word caution means. He's so eager to get his name up...."

Bronsen rose to his feet, his grey eyes flashing in hate. Reed slammed his cigar into the ashtray and threw up his hands.

"That's it! It's the last straw! I'm through playing referee for two snarling dogs. The project is closed, finished! If and when somebody can come up with a decent reason why it should be opened again, we'll consider it then. Until that time, consider the project non-existent and return to your regular jobs. And cut out the bickering and fighting—or you are both fired!"

He pulled out a fresh cigar, bit into it in disgust and dismissed the meeting by returning to the papers on his desk. Bronsen felt the anger boiling over within him and suppressed the desire to hand in his resignation on the spot. He looked for Mars and saw his thin frame out the door. He wearily passed a hand over his eyes and left the room.

Mars scowled in annoyance at Vern's whistling and silently wished the young assistant would get out of the room and let him brood in peace. He chewed the end of his pencil methodically and savagely, his features blushing pink with anger as he remembered the tirade of words exchanged with Bronsen a week ago. "Stupid, insolent, day-dreaming pup," he snarled half aloud.

Vern stopped in mid-step, eyeing him in surprise. "Huh?" he said. "Did you say something to me?"

Mars grimaced. "No. I was just talking to myself."

Vern grinned widely. "That's good. I'd sure hate to have anybody think those words were a description of me. Good old Vern, that's me. Combination office boy, slave, master of ceremonies and soothing balm for ruffled egos. That's my description of me. Master of all trades, Jack of none. Of course, I can't say what others think."

"You don't make much sense," Mars growled thickly, biting again into the pencil.

"Neither do you," Vern countered quickly. "But then again, what man does to a struggling young genius like myself?"

"Oh dry up," came the reply. "And take that drawing table into the new drafter's room."

"Oh, sure. You only need about three men to move that monster but...." He left the sentence unfinished and dragged the table from the wall. Mars smiled sympathetically, shook his head, and pointed to his bad leg when Vern indicated he could use some help.

"Such is the life of a slave," the younger man sighed and hoisting the clumsy article, headed for the door.

"Look out!" Mars suddenly yelled and jumped forward to catch a falling rocket model as the table edge glanced off it. Vern yelped in surprise, jolted backward and fell against the wall, the heavy board crashing down on his foot.

"My God, Vern. Your foot...."

The other grinned, withdrew his foot from beneath the board and pulled down his sock. "Not this baby," he flipped. "I've got cast-iron insurance. It's plastic from the ankle down, see?"

Mars stared in shock at the artificial limb and could think of nothing very brilliant to say.

"Got it in cadet school," Vern explained and then answered the question in Mars' eyes. "I was training to be a space pilot myself. Some fellows and I

decided to celebrate our graduation, got drunk and ended up in a wreck. They put me together real good, even taught me how to use the foot so no one would ever know it wasn't the real thing. It washed me out as far as the Space Corps was concerned though. I drowned my sorrows in alcohol for a couple weeks, told myself I was going to hell with myself and then decided to put what I did know to work. That's how I joined up with this outfit. Now I sit back and design the rockets my classmates have to worry about flying.... Enough of this chatter ... got to get busy. See you."

Mars turned thoughtfully back to his desk. "That kid's got only one foot," he mused soberly. He looked down at his own injured leg and savagely kicked it against the wall.

"Your leg. Your poor, crippled leg ... what a fine crutch it has been," he bitterly reproached himself. "It proves you were one of the first in space, and you won't let people forget it. You're a jealous old man. You're afraid to have someone else do what you no longer can do. You want things to stay the way they were when you got hurt, so no one else can live your dream. If time stood still, there would be no trips to new planets, no new discoveries and Mars Kenton would still be the hero of his dream."

He tried to revolt, to denounce the self-accusations. "What about Bronsen Corbow?" he asked. "Does that explain why I've fought him so hard?"

His slowly growing conscience laughed at him. "But it does. Bronsen ignores your crutch, your proof that the old way worked the best. He's concerned with the future, the future you never want to come." He buried his grey-thatched head in his hands and felt the weariness in his bones. His thoughts returned to the unsuccessful launching.

"But it was a crazy idea," he argued weakly. "It would never have worked anyway." It was a poor defense, one that faltered and failed when he finally admitted the truth: He was a jealous, bitter man, fighting anonymity.

Once more he found himself mulling over the rocket launching, probing for support to his initial decision that it wouldn't work, searching for some point to substantiate his claim. But was he really right in that decision? Had he let his hate-ridden heart rule his reasoning mind? He waded back to the beginning of Bronsen's theory. Bullets ... the test models were the bullets. Shells ... the huge rocket itself was a shell compared to the bullets. Shells

have an ojive, bourrelet, rotating band, but bullets are different. How? He stopped. He reviewed the parts in his mind, then suddenly lurched to the files and pulled out the rocket plans. He compared the ship's construction bit by bit with a shell, his mind working quickly, accurately, with a new enthusiasm....

Hours later he leaned back from his drawing table and his voice rumbled out into the quiet reaches of the empty room.

"Men will fly to the stars like a bullet," he prophesized. "Because I know why the rocket crashed."

It was dark but the light in Bronsen's office was still on. Mars pulled himself erect and turned toward Bronsen's room, then faltered. "I could just forget it," he mused. "Then the idea would be filed away. But someday...." He could not do it. The excitement was beginning to mount inside of him, pushing him forward. He took a deep breath and with a decisive shrug drew back his shoulders, standing straighter and taller than he had in fifteen long years. He strode from the room and headed down the hall.

Bronsen heard the door behind him open and close softly. He glanced up and saw who it was and returned, scowling, to his work. When Mars did not leave, he looked up again, curiosity stirring within him at the expression in the older man's face.

"Well?"

It wasn't really a question, nor an inflection denoting that he wanted to hear what Mars had to say. It was more of a compromise between physically throwing him out and grudgingly listening to what he had to say.

"I've got it, I know what happened to the ship," Mars announced quietly. "I knew it when I saw it come out of the launcher but I couldn't explain it." Bronsen returned to his papers with a snort and Mars pleaded, "I'm sorry about all those things I said. For God's sake. Listen to me!"

The tortured pleading in the man's voice made Bronsen put down the papers in surprise.

"The models worked," Mars plunged ahead. "Sure they did. But because they were small ... so much smaller than the real ship ... there was no trouble and they worked perfectly. The trouble reveals itself only as the

projectile gets larger. The nose, Bronsen. A nose band. Don't you see what I'm trying to say?"

The younger man stared in silence at the pleading ex-space pilot, before the words began to penetrate his whirling thoughts. He forgot the crash of the ship; he forgot the feel of hard teeth splitting the skin across his knuckles; he forgot the animosity that existed between them. His mind could focus on nothing but what Mars was trying to say.

"The nose of the ship is long. The only guides were on the tail at the rotating band. Think of shells. Bourrelets. The *big* shells have bourrelets ... bands around the nose that dig into the grooves and steady the front of the shell. The ship ... its front began trembling because there was nothing to guide the nose in a steady path. The more velocity the rocket had, the worse the trembling became until it threw the whole ship out of control. Don't you see? That's all that was wrong with it! It would have been perfect if it had had guide wings on the bourrelet. The guide pieces could be withdrawn when the ship is launched ... but they would have to be there in order to *get* it launched. I'm right, you know I am! That's your answer. That was the only part wrong with it!"

The enormity of Mars' words left Bronsen speechless. He looked at the suddenly joyous man before him and saw the old bitterness replaced by the rapture of his discovery. Yes, that was what had been wrong. It was the solution ... the one tiny piece that made the puzzle into an understandable picture. He paused a moment, as if trying to make a great decision, then grabbed the older man by the arm.

"Come on! Let's get it down on paper!"

The rocket lay huddled in the belly of the launching tower, her needle-like body quiet, waiting, her control panels flashing signals and instructions to her masters, her circuits buzzing with the tenseness of the seconds before blast off. The steady counting drummed through her wires, tripped relays, and her masters flipped the switches, pressed the buttons and pulled the levers that readied her for her maiden flight. Eight seconds, seven seconds.

Six seconds, five, four. The switch was jerked upward and she felt the power beginning to move in her vitals.

Three, two, one!

The driver button slammed home, her rockets roared out in ferocious birth, snarling, roaring, growing with each passing second. She settled back upon her rockets as if in protest at their screaming growth, then was forced to give ground and the ship moved up the shaft. Her rotating band and bourrelet fins dug deep into the spiraling grooves, her body began to turn ... slowly, so slowly. Then she suddenly leaped forward, her hull whirling upward; the shaft raced by in dizzy swiftness, her rockets roared louder and she raised her spinning body further. She was free! Her body hurtled up and up, her needle nose straight and true, her velocity leaping forward....

"Off rockets! Set up emergency interstellar drive for instant activation if needed. Signal in scanning screens. Activate force field and take a breather, boys. We're on our way and the blast off was perfect."

The pilot's mechanical sounding voice droned through the speaker in the moon-bound observation room and simultaneously the air was ruffled by the deep exhale of relief, the rustle of slowly relaxing bodies strung tight with the hopeful tenseness of the blast off.

Mars gazed up at the disappearing silver streak, his blue eyes intent, glistening with pride and excitement. "I never thought I'd see the day," he breathed. "Look at her, she's going straight and true. She's the most beautiful thing I ever saw."

Bronsen's face relaxed into a happy grin as the gleaming rocket hurtled up out of sight. He glanced at Mars and gave him a companionable smile.

"Even more beautiful than Mars that day? Or the old rockets?"

Mars looked slightly embarrassed and shuffled his legs into a more comfortable position. "Aw hell," he said awkwardly. "Can't you forget an old fool's ramblings? We just watched a rocket launched that's going to open up a whole new era in space travel. It was a perfect blast off and we know it'll be a perfect trip and landing."

Bronsen thoughtfully nodded his head, his grey eyes dancing.

"Tell you what," Mars continued. "I've got a bottle that I've been saving for about fifteen years. Got it when we got back from that first trip and never opened it."

Bronsen grinned and gave the old man's thin shoulder a hearty slap.

"Let's get that drink!"

*** END OF THE PROJECT GUTENBERG EBOOK CORBOW'S
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